Poco — Preserve Originals, Conceal Only

Privacy-first PII anonymisation for LLM workflows, with a fully automated multi-environment DevOps pipeline

20 / 43	8	2.4×	73:1	99.99%
PII labels viable for training	labels with zero validation instances	more tokens per character for Chinese	class imbalance (GIVENNAME : TIME)	of annotations in viable labels

The Problem

Every day, people paste contracts, medical notes, and CVs into ChatGPT. Once they do, names, addresses, and financial details are outside their control. Poco makes that safe.

How Poco Works

Poco detects personal information, replaces each value with a reversible token like [PERSON_1] or [EMAIL_1], and stores the mapping in an encrypted session vault. The redacted text is sent to any LLM, and the response is restored to its original form when it returns.

Paste text *User input document* → **Tokenise** *Reversible vault mapping* → **Restore response** *LLM response restored*

Architecture

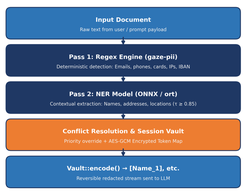


Figure 1. Two-pass hybrid PII detection engine.

DevOps Pipeline

Every code change flows through GitHub Actions, which builds the Rust backend inside the Cargo Lambda container, runs unit, integration, and browser acceptance tests, and deploys to four isolated AWS environments. Failed tests block promotion.

Commit GitHub Actions → **Build** Cargo Lambda → **Test** Unit & Playwright → **Deploy** Dev · Staging · Prod → **Monitor** CloudWatch

Live Demo

littlepoco.me

Scan to test confidentiality vault

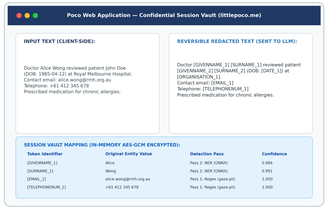


Figure 2. Poco encode view with coloured token highlighting.

Dataset Overview

Analysis of 1,636,375 multilingual documents across 30 languages, 11.9 million PII annotations, and 43 discovered PII entity classes.

1,636,375	11,887,370	43	30
Examples	Annotations	PII classes	Languages

Label Distribution

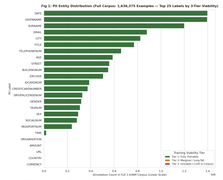


Figure 3. Top 25 labels by viability tier.

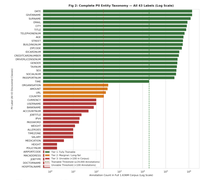


Figure 4. 43-label taxonomy (log scale).

Class Viability

Only 20 of 43 discovered labels are viable for supervised training. 4 are marginal; 19 are unviable (<100 corpus annotations, down to single occurrences). The unviable labels are handled by deterministic regex in Poco.

Language Coverage

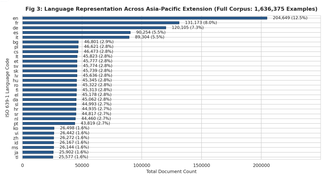


Figure 5. Language representation (30 langs).

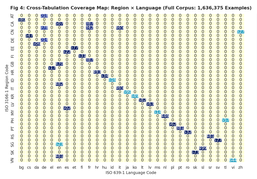


Figure 6. Region × language coverage map.

Document Lengths

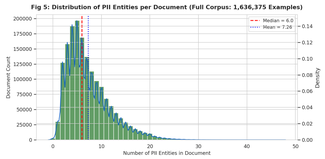


Figure 7. Entities per document distribution.

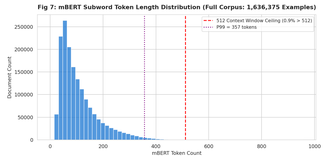


Figure 8. mBERT token lengths (512 cap).

Entity Characteristics

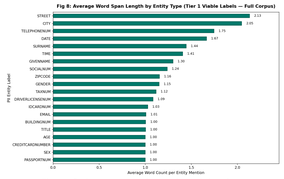


Figure 9. Average word span length by class.

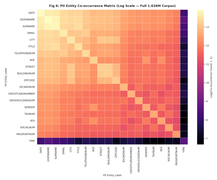


Figure 10. Entity co-occurrence matrix.

Per-Language Composition

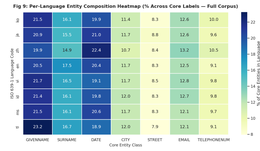


Figure 11. Per-language composition across 8 Asia-Pacific languages.

Chi-square $p < 10^{-300}$, but Cramér's V = 0.017 shows the effect is negligible — the generator produces uniform proportions.

Data Quality

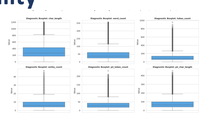


Figure 12. Pre-cleaning boxplots.

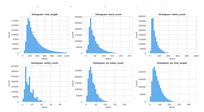


Figure 13. Pre-cleaning histograms.

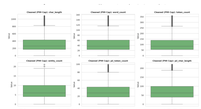


Figure 14. Winsorised boxplots (P99).

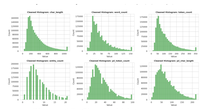


Figure 15. Cleaned histograms.

Normalisation & Correlation

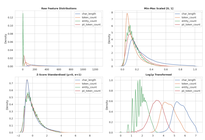


Figure 16. Raw vs MinMax vs Z-score vs Log1p.

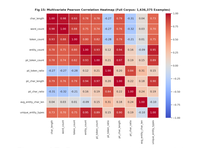


Figure 17. Pearson correlation matrix.

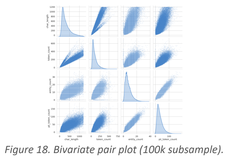


Figure 18. Bivariate pair plot (100k subsample).

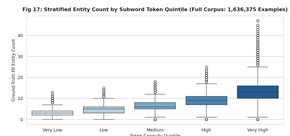


Figure 19. Entity count by capacity quintile.

Tokenizer Expansion & Split Parity

Full-corpus census confirmed that all 20 viable classes follow the 80.05% / 19.95% train/val split with $|\Delta| < 0.08\%$.

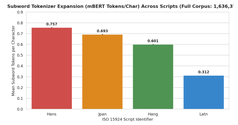


Fig 20. CJK expansion (2.4×).

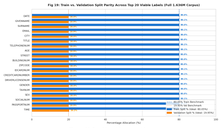


Fig 21. Split parity.



Fig 22. Validation blindness.

Repository github.com/Schnitze1/Poco

Live Demo littlepoco.me

Full EDA Report Scan for full-corpus analysis